

ROYALE BINGHATTI - DUBAI

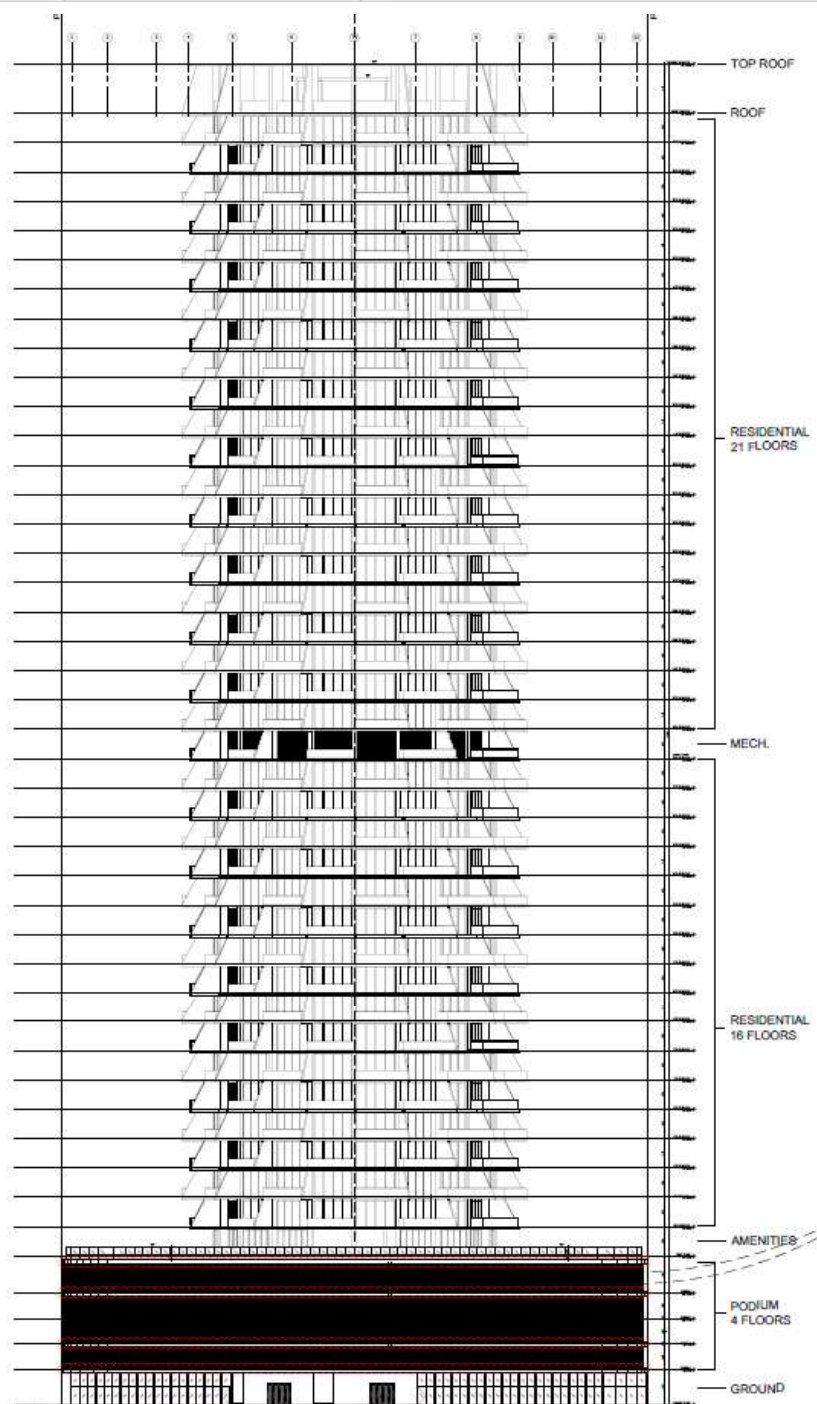
TECHNICAL STUDY PROPOSAL



PROJECT DATA

Request Details

Height above DMD level (meters)	Height above ground level (meters) *	Position
182.7	174.3	Binghatti Royale



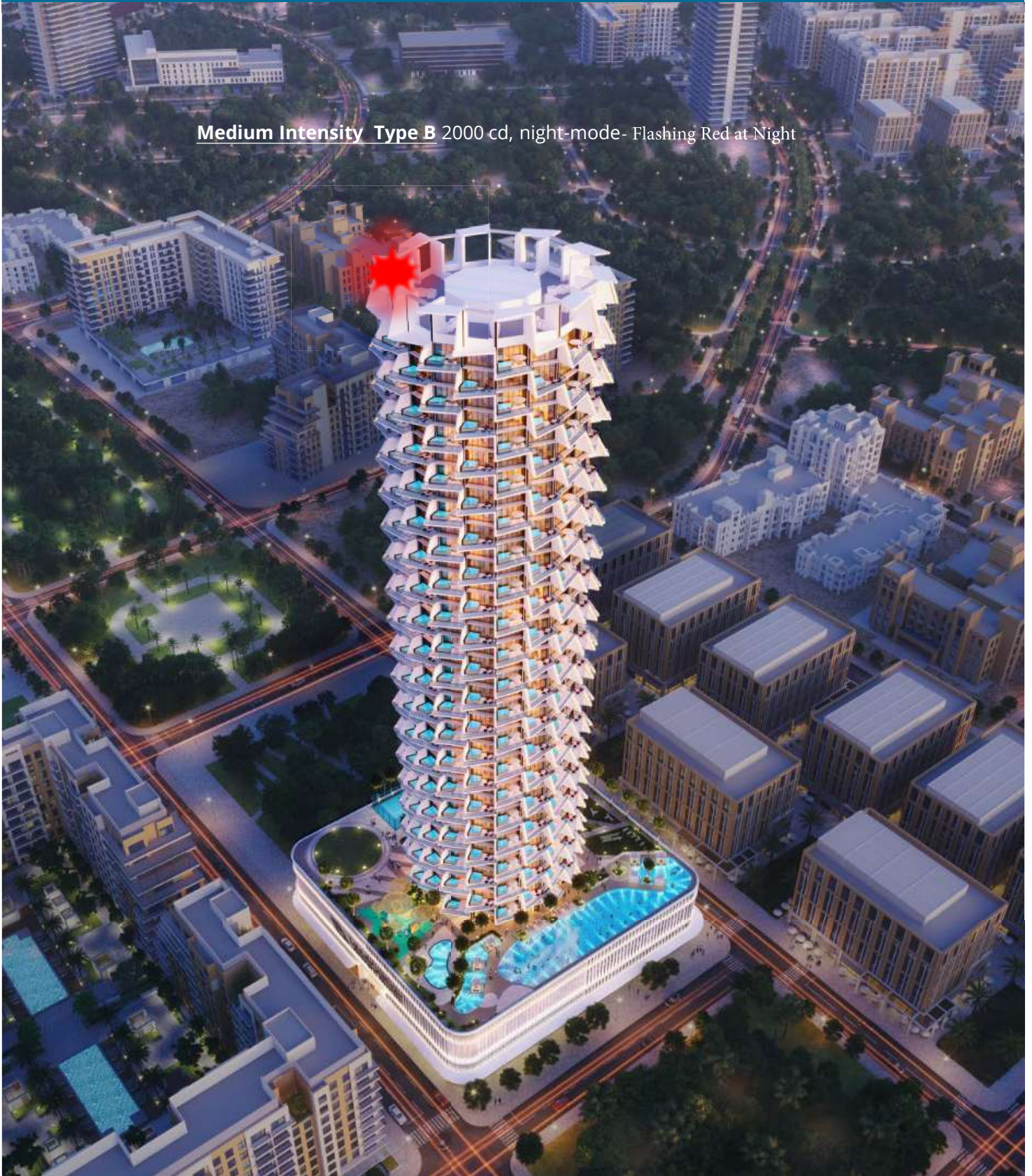
Light Intensity definition according to ICAO and FAA

INTENSITY	COLOR	Type		Flashes per minute	NIGHT (Cd)	DAY (Cd)	TWILIGHT (Cd)
		ICAO	FAA				
LOW	Red	A		Steady	> 10	-	
		B	L-810	Steady	>32,5		
MEDIUM	Red	B	L-864	20-40 (FAA) 20-60 (OACI)	2,000 ± 25%	-	
		C		Steady	2,000 ± 25%		
	White	A	L-865	40 (FAA) 20-60 (OACI)	2,000 ± 25%	20,000 ± 25%	20,000 ± 25%
			L-866	60 (FAA) 20-60 (OACI)	2,000 ± 25%	20,000 ± 25%	20,000 ± 25%
HIGH	White	A	L-856	40	2,000 ± 25%	200,000 ± 25%	20,000 ± 25%
		B	L-857	40	2,000 ± 25%	100,000 ± 25%	20,000 ± 25%

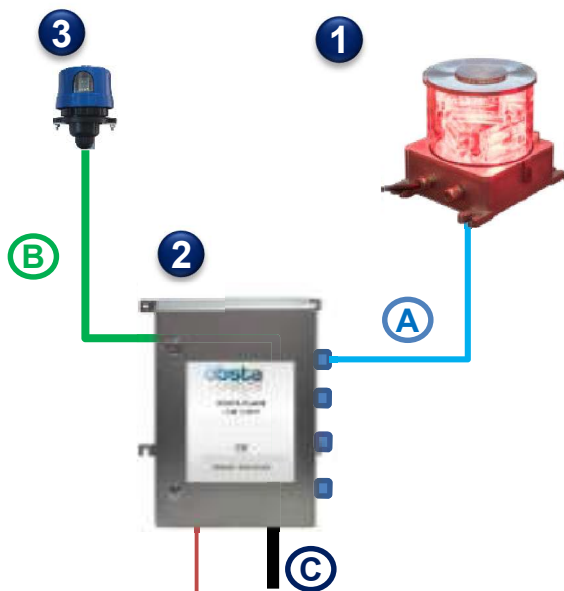
OBSTA PROPOSAL

ELEVATION VIEW - **VIEW OF THE BUILDING AT NIGHT**

Medium Intensity Type B 2000 cd, night-mode- Flashing Red at Night



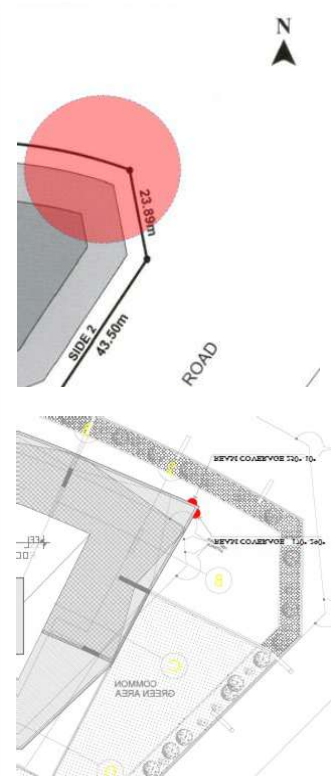
SINGLE LINE DIAGRAM



Sl.	Description	Part Number	Reference
①	MI Type C	113790RI-240-G	OFC-RI-240-G
②	Control Cabinet	113176-240	OFC-CTR-240
③	Photocell	100757	PHOTOCELL DC
Ⓐ	AW1 Cable	5G1,5sqmm	HELUKABEL
Ⓑ	Data Cable	3G1,5sqmm	HELUKABEL
Ⓒ	AC Power Cable	3C 2,5 or 4mm	Local single phase supply
Ⓓ	BMS Cable	-	-

MEDIUM INTENSITY

BINGHATTI, DUBAI UAE



Commissioned June 2022. OFC-RR-240 was configured to type C medium-intensity connected to a controller with a photocell.

MEDIUM INTENSITY

SEA WORLD - ABU DHABI, UAE



Ref: ADAC/NOC/01/20214416
Date: 10 Jan 2021

Subject: **ADIA – Aircraft Warning Lights No Objection Certificate (NOC)**
Sea World Abu Dhabi, Sector YS6 Plot C1, Yas Island Abu Dhabi

Application No. 4416 dated 07 Jan 2021
Dear Sir,

We refer to the above referenced letter that was forwarded to us for technical analysis regarding Aircraft warning lights design approval.

ADAC have studied the design and specifications for aircraft warning lights at the site given above and pleased to confirm our 'No Objection' for the following height.

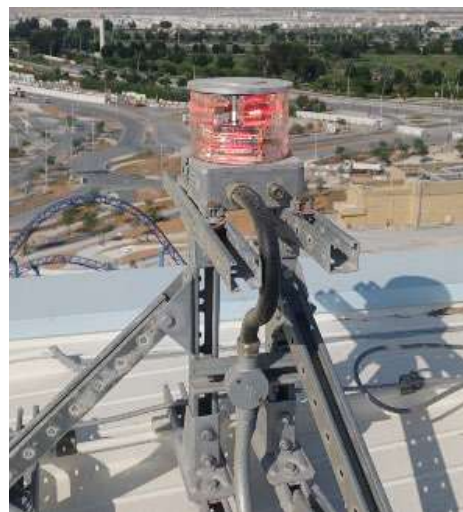
A total number of **17 medium intensity obstacle lights LED type C Red fixed (Model: OFC-RR-240-RG) Make: M/s Obsta-France**

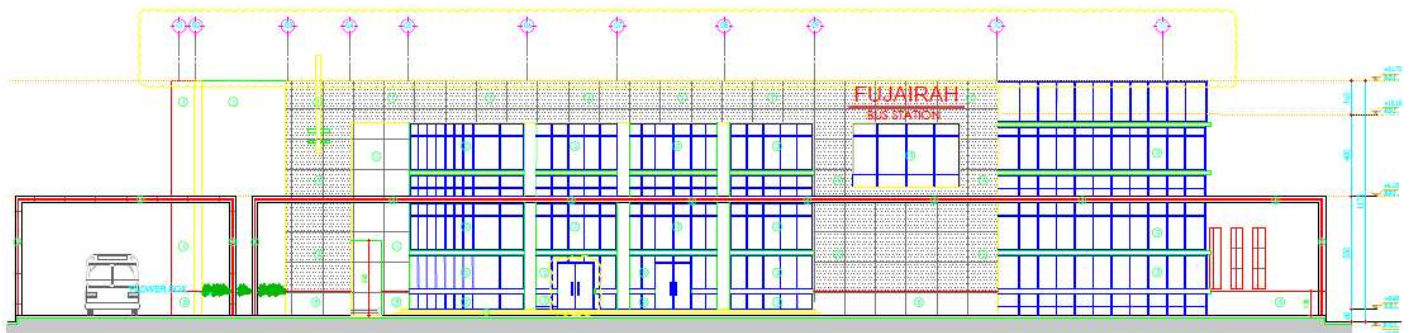
Trusting this meets your requirements and assuring you of our continued attention.

Yours Sincerely,


Naeser Juma
General Manager ANS & AGL

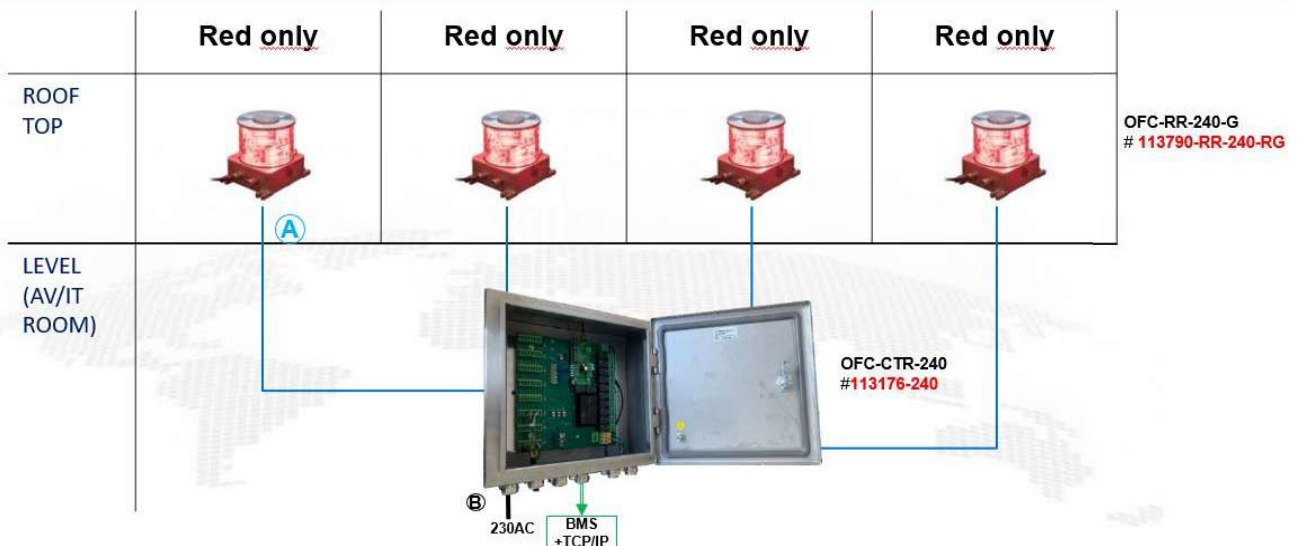
cc: Asim Rizwan – ADAC ANS





3. OBSTA PROPOSAL

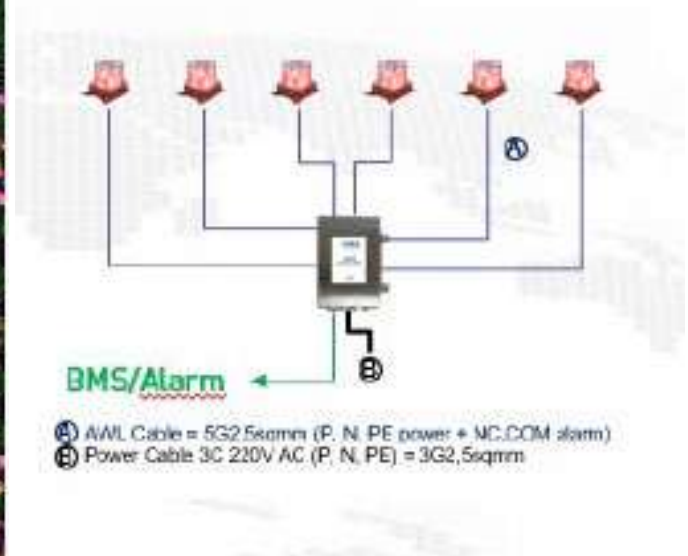
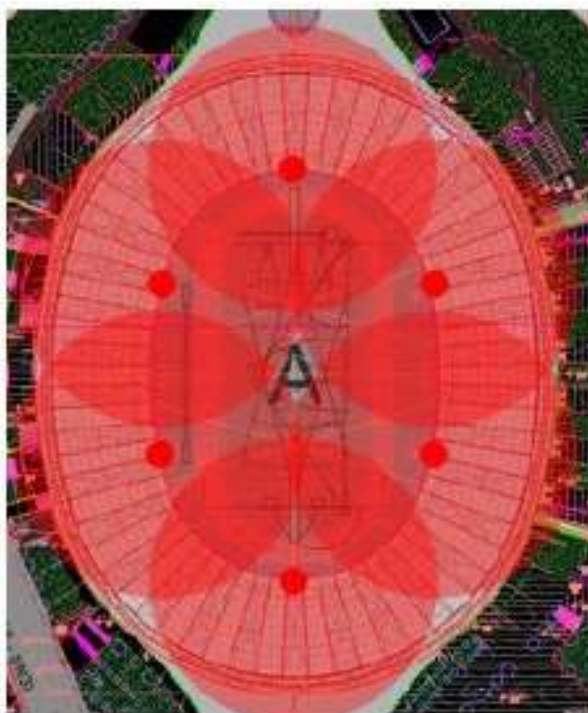
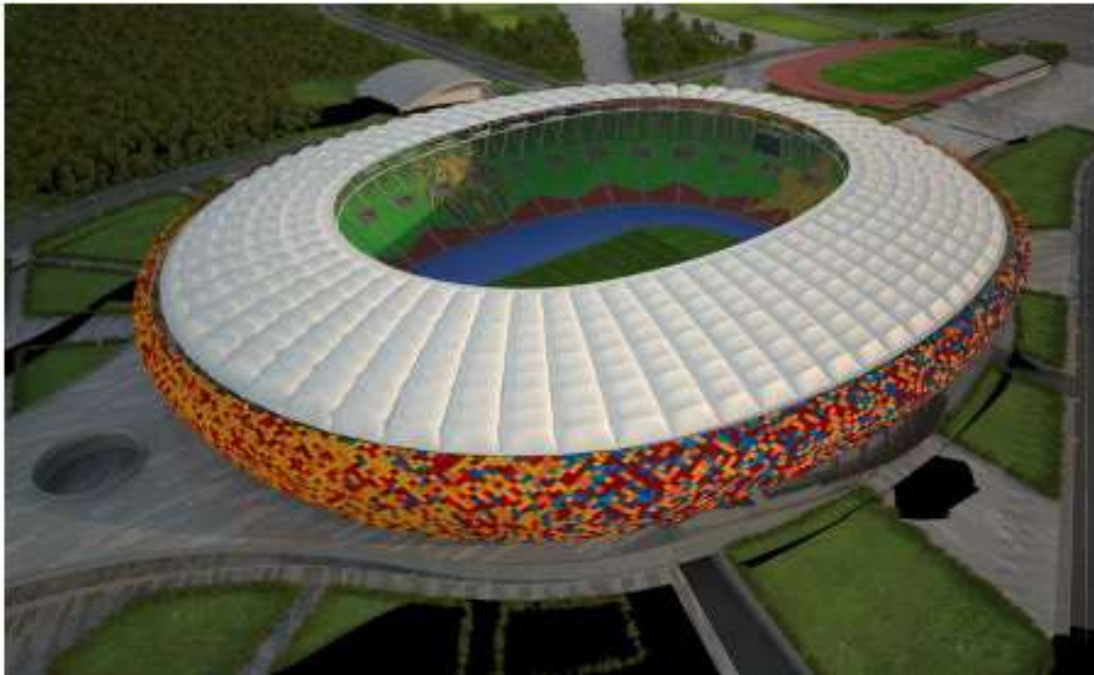
2. SINGLE LINE DRAWING



(A) AWL Cable 5G1,5mm² (P, N, PE, 2 x alarm)
 (B) Power Cable 3C 220V AC (P, N, PE)

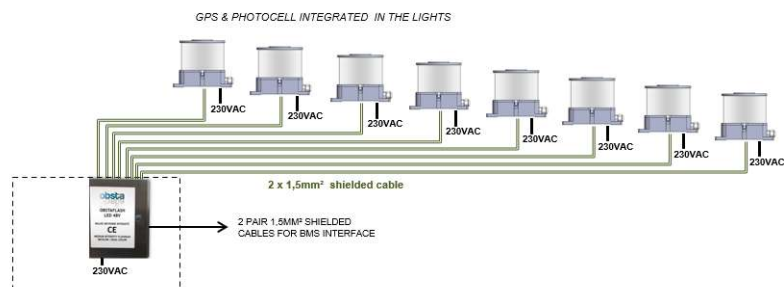
- ☑ All lights equipped with Photocell and GPS module for full wireless flash synchronization
- ☑ All lights equipped with Power Supply module for operation and protection

MEDIUM INTENSITY OLEMBE STADIUM, CAMEROON



MEDIUM INTENSITY

AL THUMAMA STADIUM, QATAR



MEDIUM INTENSITY

OBSTAFASH COMPACT RED INTEGRATED AND REDUNDANT

EAP/EHV Engineering & Projects Extra High Voltage Projects Department REVIEW OF CONTRACTOR'S SUBMITTAL			
<i>(Use for Civil, ME or Mechanical, E for Electrical, S for Safety, CM for Commissioning, SW for Safety)</i>			
Submittal No.	PE18-E-181.R.01	Date:	29-May-18
PROJECT TITLE :	CONSTRUCTION OF REC-330KV/33KV XS & ASSOCIATED BKKV DC CHD IN IRAS AL KHUAR		
CONTRACT NA# :	REC 999-C-05 PTS : I. PFS 140300 S. PFS 140300	JO NO# :	RC 099-C-05
To : EAP / EHV PROJECTS DEPARTMENT Room # 2-9006, SEC-EQA HQ, Dammam	From : Affanar Co. "P.O.BOX 13103, DAMMAM 31460, KSA"		
ACCEPTANCE OF THE FOLLOWING SUBMITTAL IS HEREBY REQUESTED: Check the box: ☐ Drawings ☒ Materials ☐ Test Reports ☐ Others (Specify) _____			
FILL-IN THE INFORMATION BELOW:			
Description	(Materials Submittal) for Obstruction Removals and Warning Lights		
Reference Specification	1. PFS-14E/D306 OHTL		
Manufacturer/Supplier	M/s. CBSTA		
Vendor Address :			
Expected Delivery Time (Attachment's)	Completion statement of Rec'd - Attached Document + CD		
COMPLYING WITH SCOPE OF WORK AND TECHNICAL SPECIFICATIONS? ☒ YES ☐ NO ("NO" INDICATE DEVIATIONS : (Provide justification and attach supporting documents))			
Contractor : Affanar Co.		[CLIENT : Saudi Electricity Company - EQA]	
Submitted by : Signature	Received by: Signature		
Name : Sami Abdul Aza	Name : Syed Shafiq Ali		
Project Manager	Project Engineer, EHV Projects Department - EQA		
Date : 29-May-18	Date : 28/5/2018		
FOR SEC-EQA USE ONLY			
SEC REVIEWED THE ABOVE SUBMITTAL AND FOUND IT:		REMARKS :	
☒ ACCEPTABLE ☐ ACCEPTABLE AS NOTED ☐ NOT ACCEPTABLE (RESUBMIT) ☐ PROVIDE ADDITIONAL INFORMATION ☐ SEE ATTACHED COMMENTS		No further comments.	
Signature : Hamed Hussain		Please use extra sheets if needed.	
Name : Hamed Hussain		Received by: Signature	
Date : 11-06-2018		(Contrictor) Date : 12/6/2018	
SEC-EQA Internal References		Date : _____	
18-1870		EHS/EP/LDP: 297-0.HH: 10-06-2018	

NOTE: AFFANAR SHALL RELEASE THE CONTRACTOR FROM ALL RESPONSIBILITIES IN PERFORMING THE WORK AS SHOWN CONFORMANCE WITH CONTRACT, SCOPE OF WORK AND TECHNICAL SPECIFICATIONS.

Mr. Tawfik M. Almetwally (General Supervisor) International Division


ENTECH ENGINEERING & INVESTMENT COMPANY S.A.O.G.
 ENTEC ENGINEERING & INVESTMENT COMPANY S.A.O.G.
 ENTEC ENGINEERING & INVESTMENT COMPANY S.A.O.G.

No : EPE-ONEIC-85/2014-436 Date:02-03-2020

To : M/s. Oman National Engineering & Investment Company S.A.O.G
 PO Box 139,Rawi, Postal Code 112, Sultanate of Oman

Fax No. : 23996499

Attn. : Mr. Arumugamsamy, Project Manager

Project : Construction of 120kV Merlat GS with 2x50MVA and Associated 132kV
 OHL from Ashbur GS to Merlat GS (Tender No.85/2018)

Sub : OHL:MS for Tower obstruction lights, warning spheres -Approved
 Reference : 1/DTN-EPE-85/2018-MSAS-OHL-01 dated 19-02-2020

Dear Sir,

With reference to the above, please see the following:

S/N.	Title of the document	Rev	Status
1	MOS-85/2018-MSAS-OHL-01 MS for Tower and Conductor Warning Lights and Aircraft warning spheres	0	Approved

Comments : All the technical requirements and dimensions shall be coordinated and confirmed with OPEC. The manufacturer please confirm and submit a compliance statement only

This is for your information and necessary action.
 Regards,
 Yours faithfully,


A.SATHIYAKUMAR
 Project Manager

Encl. : as above
 CC : OPEC, Senior Manager (Projects)

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E-Mail : entech@enteconline.com Website : www.enteconline.com

obsta™
RELIABILITY IN
OBSTACLE
AVOIDANCE

4. OBSTACLE FLASH R&B INTEGRATED DATASHEET

Medium Intensity type B or C and
FAA L-B&A compliant with 2 LED
circuits in active redundancy
(one main and one standby).

Self-contained, compact and light
system with integrated power supply
and cast moulded fixing holes

PHOTOMETRIC FEATURES

- Effective light output of 30 or 40candelas
- Vertical beam spread of 30°
- Horizontal beam spread 30°
- 4 x OBSTA In-house designed optics
(see standard drawing)

ELECTRICAL FEATURES

- 2 x red LED circuits in active redundancy
- Adjustable flash (20 to 60 form or steady burn)
- Alarm in case of LED circuits or power failure
- Power supply 12, 24, 48VDC or 120/230VAC
- Average power consumption of 3.27W (220V FPM) and 3.2W (24V battery mode), class 1 & 2
- AC or AEEC surge protection included
- Modular base design

Mechanical Features

- Pure hard glass cover (no acrylic plastic)
- Aluminium base
- Ultra compact optics and base design
- Cast moulded fixing holes in base
- Connection by cable
- Base and lamp
- Fastening with metal clamp

Approved
Fidelmar Dps - CNIL 64

CUSTOMER BENEFITS

- ✓ ICAO and FAA compliance on all criteria
- ✓ Self built-up and repairable
- ✓ Specifically designed for this application
- ✓ Optimized beam

CUSTOMER BENEFITS

- ✓ Self built-up and repairable
- ✓ To comply with any Civil Aviation requirements
- ✓ 100% reliability and control
- ✓ All alternatives to connect to grid or solar kit
- ✓ Reduced consumption and higher durability
- ✓ Improves robustness in hard environment
- ✓ Optional interface for GPS and photocast

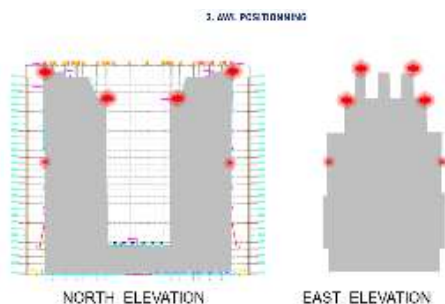
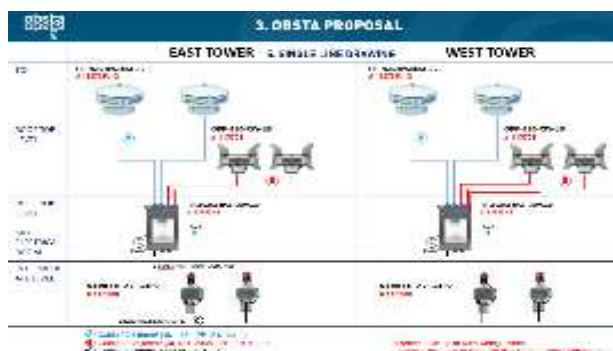
CUSTOMER BENEFITS

- ✓ No fading from ageing, sand erosion and frost
- ✓ Operation up to 45°C
- ✓ 220mm x 200mm x 200mm and 400g weight
- ✓ Quick installation on top of towers
- ✓ Quick activation
- ✓ No maintenance and no special tools to install

Same product offered is approved and in continuous use for 5 years in by far harder conditions.
On top of High Voltage transmission lines supporting towers.

On all coastal and salty area and through all deserts in MENA area up to 60°C.

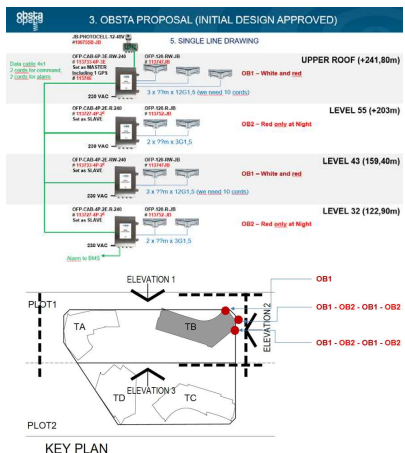
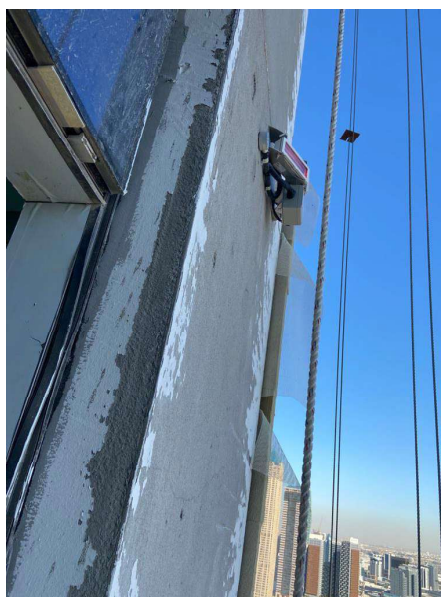




We did comply with the twin design of the building by mixing our 360° versions on top and 180° on intermediate levels



Perfect and easy installation here on vertical facade, on concrete and glass.
We had to strictly respect DCAA requirement of specific beam angles.
Our 120° certified and compliant products offer this flexibly without the need to had extra bracketry and shielding.



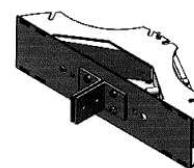
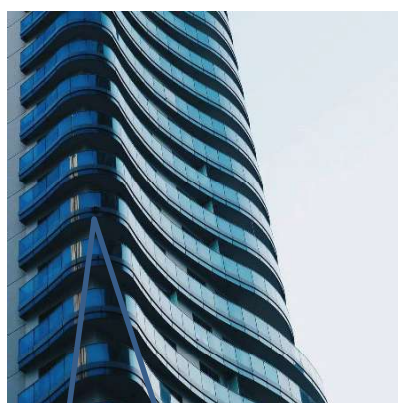
VERCASE TOWER BEIRUT CENTRAL (INSTALLED IN 2015)

SAME CERTIFIED SYSTEM IS IN USE IN ALL
CONDITIONS

Our aluminium and stainless steel Flashheads can be integrated into the design



***DAMAC Tower with interiors by Versace Home
Beirut Central (installed in 2015)***



2. SOME REFERENCES IN ALL CONTINENT AND CONDITIONS



ETISALAT Towers
Abu Dhabi



TV antenna above 200m
Texas, USA



More than 1000
OBSTAFLASH
accumulated from 2013 on
power lines projects in all
climates from **Russia** to
Saudi Arabia, Oman,
Egypt...



Four Seasons
Hotel, **Abu Dhabi**



Damac
Versace
Tower, **Beirut,**
Lebanon



TOTAL OIL &
GAS HQ
LAGOS,
NIGERIA

Diegem
Cathedrale
Zaventem Airport,
Belgium



Factory Chimney
France



Thermal Power
Plant
Tunisia